# FARSuN Historical Sunspot Database

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The FARSuN historical sunspot database consolidates raw sunspot observations (1610–1980) from multiple legacy sources into a single, queryable system that follows Virtual Observatory (VO) standards. It harmonizes independent encodings (e.g., Mitteilungen, Wolf Institute tables, Shreya's FARSuN DB, Stephen Fay's variant) and augments them with observer/observatory/instrument metadata, provenance, and QC flags to support scientific reuse and the reconstruction of the Sunspot Number (SN) V3. The core content is daily/group/spot counts (NG, NS, Wolf number components) with optional positions/areas where available, mapped to observers, observatories, instruments, and bibliographic rubrics (Mitteilungen IDs, pages/notes). It captures both table-based sources and drawing-derived extractions. The database is the central integration layer: it makes disparate historical observations Findable, Accessible, Interoperable, and Reusable (FAIR) via standard VO tooling; enables consistent QC; and provides the authoritative input stream to SN V3 stitching/calibration workflows. It also supports new products (e.g., GN; higher cadence daily numbers) by ensuring traceable, versioned provenance.

# 1 A consolidated, QC'd, VO-exposed database is essential to:

- Quantitatively compare observers and epochs to derive robust k-factors and correct systematic biases (e.g., pre-Wolfer vs. post-Waldmeier eras).
- Reconstruct SN V3 with explicit provenance, uncertainty, and versioning.
- Enable external tools (heliophysics, climate studies) to query the same canonical, machine-actionable tables.

# 2 What the data look like (schema snapshot)

- **Core table:** `observations` (date, NG, NS, Wolf components/total, optional positions/areas, method/conditions, links to observer/observatory/instrument/source/rubric, provenance, QC flags).
- **Entities:** `observers`, `observatories`, `instruments`, `sources`, `rubrics`.
- **Control & traceability:** alias and legacy ID mapping, `provenance`, `qc_flags`, `audits`.
- **Assets:** links to scans/snippets with OCR/HTR confidence.

# 3 Current status (high level)

- **Legacy sets staged** in separate schemas for deterministic comparison: initial Mitteilungen, Shreya Mitteilungen, Stephen variant, Wolf Institute dump, Shreya FARSuN.
- **Web interfaces loaded** for curator review; mapping tables in place.
- **New extractions underway:** Adams (drawings), Gruithuisen (manuscripts); Zürich tables extracted by students.

# 4 Quality control (QC) highlights

- Structured **comparison views** across legacy vs. canonical tables.
- Automated checks: date integrity, **NG NS**, missing observer names, **interpolated values**, duplicate candidates.
- Documented corrections: observer misattributions (e.g., Williston Young; Main vs. Airy).
- **Observer–year–observatory** cross-correlation tests in curator tooling.

## 5 Digitization & metadata

- Mitteilungen tables digitized and transferred; annotations and corrections applied.
- Drawing extraction stack evaluated (DIGISUN, NSO tools, Arlt code); API-based snippet workflow for manuscripts.
- Metadata build-out for **EPNcore alignment** (observer, instrument, observatory, source/rubric); Zürich tables metadata encoding ongoing.

## 6 Accessibility

- Target exposure via **EPN-TAP** on ROB's TAP server (already validated with USET tables).
- Results queryable with **ADQL**; delivery as **VOTable**; documented release and versioning policy.

## 7 Roadmap toward SN V3

- **Finalize canonical schema** (conditions vocab, instrument/observatory validity ranges, mandatory provenance/audits).
- **Complete legacy ID crosswalks** and alias disambiguation; close mapping gaps.
- **Scale QC automation** (validator suite + dashboards) and uncertainty model (observer variance, method/conditions).
- **Observer intercalibration & stitching** per historical regime; publish alpha/beta V3 comparisons.
- **VO exposure** of `epn_core` view + example ADQL notebooks; register service.
- **Extraction scale-up** (Adams, Gruithuisen, Bernaerts, Herschel, Pastorff, Soemmering, Rost); integrate HTR confidence into QC.

## 8 Risks & follow-ups

- Residual legacy inconsistencies and incomplete aliasing/metadata.
- Variable availability and accuracy of positions/areas across sources.
- Need **EPNcore extensions** for sunspot-specific fields (Wolf components, QC ranks). **Mitigations:** curated mapping closure, authority control, QC validator expansion, documented EPNcore extensions.